

FEA REPORT

IMPERIAL COLLEGE LONDON

DEPARTMENT OF DESIGN ENGINEERING

Prosthetic Running Leg

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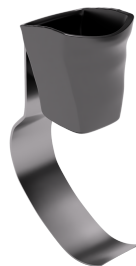


Figure 1: Prosthetic running blade model

Imperial College London

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1 Introduction

The goal of this report is to use finite element analysis to understand the vibrational and fatigue properties of a prosthetic running leg.

1.1 Method

This report covers 2 designs, the first design uses CFRP, a composite material commonly used in blades [1]. Due to a lack of detail on fibre orientation and layup information, and the complexity of representing anisotropic behaviour in a curved geometry, the CFRP was modelled as an isotropic material. The second design focuses on fixing the flaws highlighted in the original design. Next there will be a comparison between the CFRP and an aluminium alloy. All tests are performed in Ansys Mechanical.

In terms of device longevity, CFRP is assessed using a factor of safety approach, as its fatigue behaviour is complex and less predictable due to its brittle nature. In contrast, the aluminium alloy is evaluated using fatigue life analysis based on its S-N curve.

For calculating the safety factor, Von-Mises equivalent stress was used in this case because the material has been modelled as isotropic and it combines normal and shear stresses in a multi-axial loading state. Maximum principle stress was also considered as it is the typical stress used in brittle materials, however, when applied the results centred around the fixed support and were therefore seen as less representative of blades true behaviour. Since Von-Mises stress showcased typical blade behaviour and allowed for consistency between material tests since the fatigue test also used Von-Mises it was seen as the better choice.

For the vibrational study, both designs and materials use the fundamental frequency.

1.2 Criteria

To classify the design as successful, it must achieve a fundamental frequency greater than 50 Hz (which is safely above the typical excitation range of running) and either a factor of safety exceeding 3 or a fatigue life greater than 1,000,000 cycles.

The fundamental frequency of the blade can be approximated by

$$f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}, \quad (1)$$

where k is the structural stiffness and m is the effective mass. This shows that increases in stiffness will increase the natural frequency.

The stiffness of the blade is related to its bending stiffness, given by:

$$k \propto EI, \quad (2)$$

where E is the Young's modulus and I is the second moment of area for a rectangular cross-section:

$$I = \frac{bh^3}{12}, \quad (3)$$

where h is the thickness and b is the width. This relationship highlights that stiffness is sensitive to thickness, as $I \propto h^3$.

Excessive stiffness would reduce the blade's ability to elastically deform and store energy, compromising its intended function as an energy-storing and returning system. Therefore, the design must balance the above properties to maintain sufficient flexibility.

2 First Design

As stated above thickness has the largest impact on stiffness, therefore the goal of the first design is to experiment with how to balance thickness with flexibility.

2.1 Geometry

The model blade was created using the form tool following a sketch. The total blade height is 417 mm and the blade width is 100 mm, 130 mm of the height consists of the connection the socket, the remaining 287 mm is the actual blade. The thickness varies throughout the blade, with 8 mm at the socket, 5 mm at the midpoint of the curve, and 6 mm at the contact point. The main goal of this design was to allow enough flexibility for elastic energy storage under low-frequency loading, while maintaining sufficient stiffness to ensure a high natural frequency and avoid resonance.

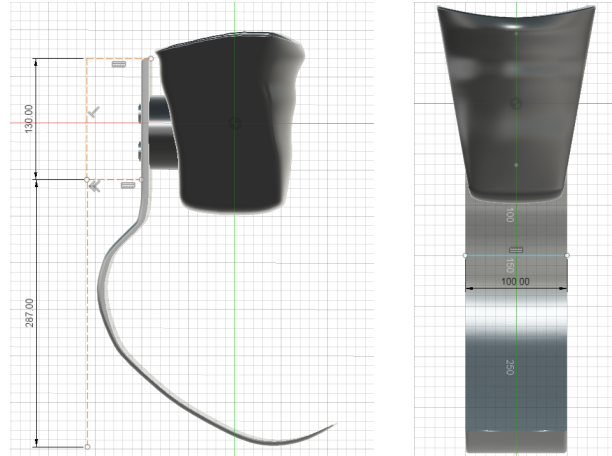


Figure 2: Height and width views of the first blade geometry.

2.2 Boundary Conditions

In Ansys mechanical the bolts and socket were suppressed, next the bolt holes were assigned as the fixed supports and a force was assigned at the lowest point of the blade with a normal reaction force of 2100N and a frictional force of 500N opposing the direction of motion.

2.3 Meshing

Before meshing took place, the model was de-featured to prevent issues with mesh focusing. There were 4 stages to the mesh refinement process: first a high element size mesh was made (10mm) to check if the model was behaving as intended and

to discover the areas of stress concentration to understand where the mesh needed to be focused. Next the mesh element size was altered in a range from 10 mm to 1 mm to discover where the max Von-Mises equivalent stress converged. After the results converged and mesh quality was reviewed, the next step was creating areas of focused sizing. Finally furrow checks were done on the details of the mesh quality.

2.3.1 Interval led Refinement

As seen in table 1 and figure 3 the results started to converge at 5mm with another small increase and convergence of stress at 2.5-3 mm. The graph does not include results lower than 2.5mm because there was a viewed spike in stress that was discovered to be caused by a drop in mesh quality and any lower than 1.75mm was outside of the student license for solving.

Element Size (mm)	Stress (MPa)	Stress % Change
10	299.9	
9	301.4	0.5
8	300.3	0.36
7	308.1	2.6
6	303.5	1.5
5	303	0.16
4	303.5	0.17
3	304.4	0.3
2.5	304.2	0.066

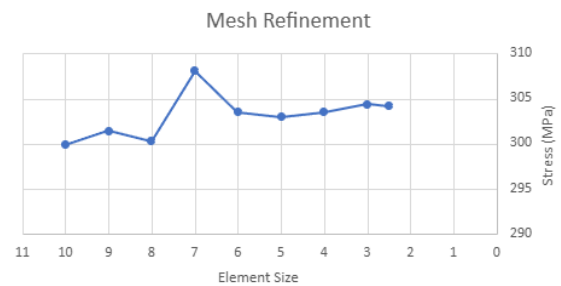


Table 1: Mesh refinement results.

Figure 3: Convergence of stress with element size.

2.3.2 Quality Check

The minimum element quality, skewness and aspect ratio came out alarmingly low or high (0.037, 1, 83.179).

Quality Criterion	Warning Limit	Error (Failure) Limit	Worst
Max Aspect Ratio	Default (5)	Default (1000)	83.179
Min Element Quality	Default (0.05)	Default (5e-04)	0.037
Min Jacobian Ratio (Corner Nodes)	Default (0.05)	Default (0.025)	0.037
Min Jacobian Ratio (Gauss Points)	Default (0.05)	Default (0.025)	0.36
Max Element Edge Length	Default (0.248 m)	Default (0.497 m)	0.011 m

Table 2: Mesh quality results.

Further analysis into the contours and graphs showcased that the issue centred around the tip off the blade which when looking at the stress contour (figure 10) it can be seen that it is not a place of significance within the mesh while the rest of the mesh is high quality.

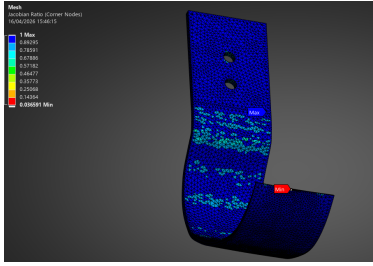


Figure 4: Jacobian ratio contour.

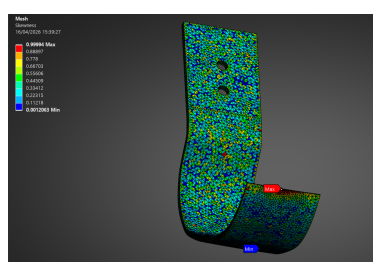


Figure 5: Skewness contour.

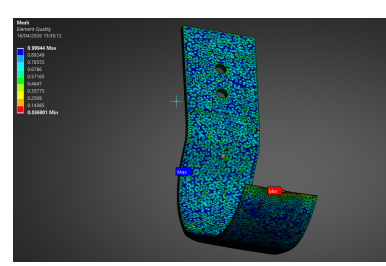


Figure 6: Element quality contour.

2.3.3 Sizing Focus

A focus with a 2.4 mm element size was added to the mesh. This sizing behaved oddly becoming much smaller than the global despite the 0.1 mm difference, any further decreases caused the mesh to be outside of the license budget.

Max Aspect Ratio	Default (5)	Default (1000)	112.5
Min Element Quality	Default (0.05)	Default (5e-04)	0.04
Min Jacobian Ratio (Corner Nodes)	Default (0.05)	Default (0.025)	0.106
Min Jacobian Ratio (Gauss Points)	Default (0.05)	Default (0.025)	0.397
Max Element Edge Length	Default (0.248 m)	Default (0.497 m)	9.8e-03 m

Table 3: Focus mesh quality results.

While adding a focus did not affect maximum stress convergence it did affect average stress and mesh quality (table 3). The Jacobian ratio was pushed into the green and issues can be seen to be visibly reduced on the contour. While the aspect ratio does get slightly worse it can be viewed to again only focus on the tip of the mesh, also the overall mesh element quality can be seen to have also improved with focusing. The focused contours can be seen bellow (Figure 7, 8, 9).

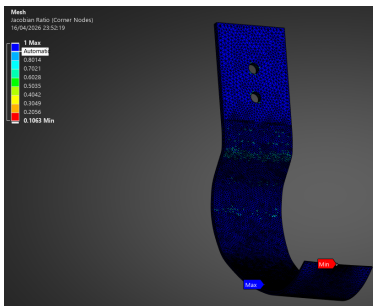


Figure 7: Jacobian ratio contour.

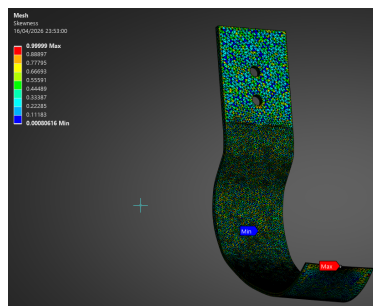


Figure 8: Skewness contour.

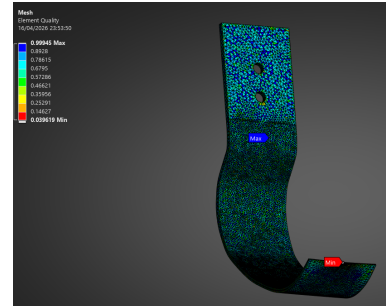


Figure 9: Element quality contour.

2.4 Results

The max Von-Mises equivalent stress was equal to 304.64 MPa, the total deformation was equal to 5.35 cm and the fundamental frequency sits at 51.47 Hz.

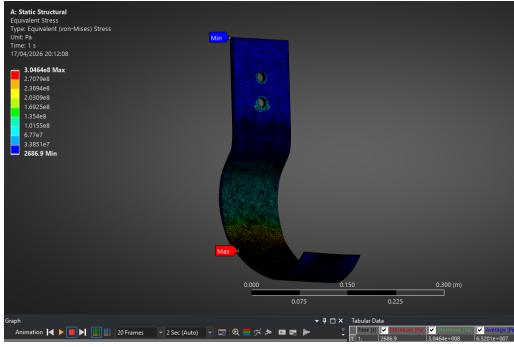


Figure 10: Von Mises equivalent stress contour.

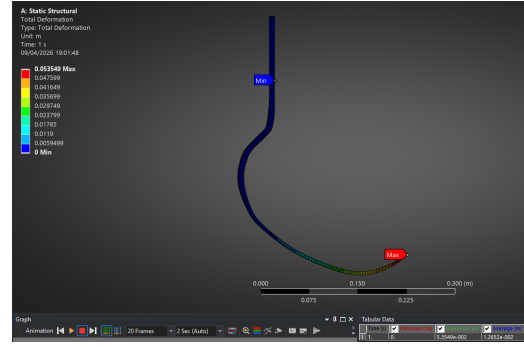


Figure 11: Deformation contour.

2.4.1 Discussion

The first design satisfies the fundamental frequency requirement of 50 Hz. The factor of safety (FoS) was calculated as the ratio of the ultimate tensile strength of the material to the maximum equivalent stress (von Mises). The failure stress was 600 MPa, the resulting FoS is approximately 2. This falls below the value set in the criteria (3), indicating that the design is insufficiently robust. The results suggest that reinforcement is needed, particularly in regions of high stress concentration, where increasing thickness would increase local stiffness and overall structural integrity. At the same time, the average thickness should not be significantly increased, as this would increase structural stiffness too much and shift the fundamental frequency further above 50 Hz, potentially compromising the required dynamic response of the blade.

In terms of stress distribution, the contours show the highest stress is on the edges of the blade rather than bands across the blade which would be the expected result as seen in areas with medium stress concentration. This suggests that the curvature of the blade is highly effective at distributing stress throughout the geometry.

3 Second Design

Since the geometry has not changed significantly, the mesh size was kept the same at 2.5 mm global element size and 2.4 mm focus. Maintaining a consistent mesh size allows for direct comparison between designs, as variations in mesh density can influence how stress gradients and peak stresses are captured. By keeping the mesh size unchanged, differences in results can be attributed solely to geometric differences.

Meshing quality checks were then performed and it resulted in similar values to the original focus mesh.



Figure 12: Second design mesh.

3.1 Geometry

The aim of the geometric modification was to increase the factor of safety to 3 or above. This was achieved by altering the tapering profile to a more gradual transition from 8 cm to 5.5 cm. As a result, the regions of peak stress are locally thicker, reducing stress concentrations without significantly increasing the overall average thickness of the blade.

In addition, the curvature of the blade was modified by extending the curve slightly further outward and closer to the ground. This adjustment was intended to improve the load distribution along the blade, promote a more uniform stress profile, and contribute to an additional increase in the factor of safety.

3.2 Results

3.2.1 CFRP

The total deformation is equal to 2.6 cm, the maximum Von-Mises equivalent stress is equal to 188.88 MPa and the fundamental frequency is 54.986 Hz. These results act as a sanity check, all values are within material limits and the frequency is reasonably above the design requirement threshold. Results can be seen in figures 13 and 14

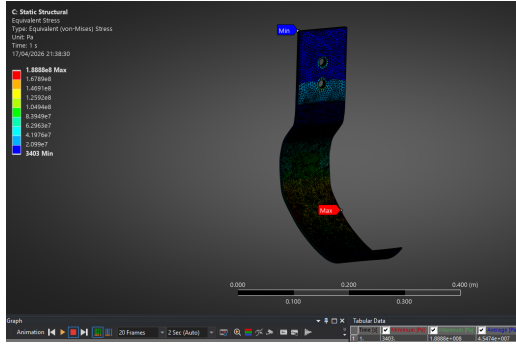


Figure 13: CFRP Von Mises equivalent stress contour.

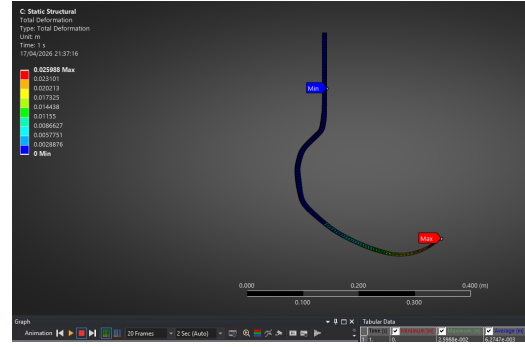


Figure 14: CFRP Deformation contour.

3.2.2 Aluminium

The aluminium used was the basic aluminium alloy in the Ansys database. The total deformation is equal to 2.53 cm this can be seen in figure 15 and the fundamental frequency was 43.055 Hz.

The minimum life cycle of the blade was calculated via the Ansys fatigue tool. This tool uses the materials S-N curve which is the relationship between stress and the number of cycles until failure. The fatigue tool was zero based to simulate that stress is applied in impacts and the runners blade when lifted experiences near zero load. The Gerber correction was used because it is less conservative than other corrections such as Goodman or Soderberg corrections. It also assumes a parabolic relationship between mean stress and alternating stress which is better suited for ductile materials such as aluminium. Results for the minimum life cycle came out at approximately 19 million cycles as seen in figure 16.

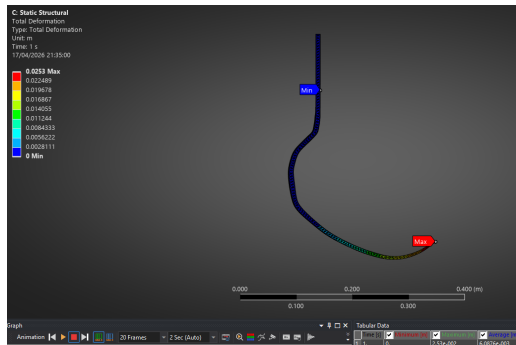


Figure 15: Aluminium Deformation contour.

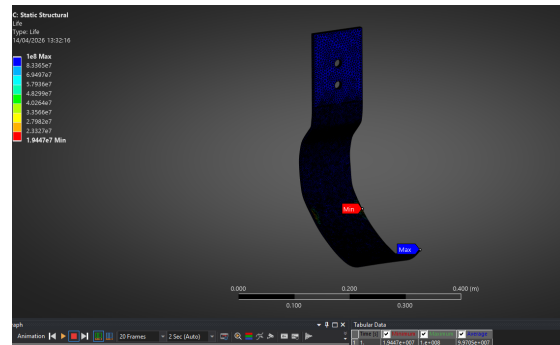


Figure 16: Life cycle contour.

3.2.3 Discussion

The second design iteration was successful, With CFRP it both sits in the fundamental frequency range and has a factor of safety of approximately 3.2.

As predicted, the blade became less flexible due to the increase in local thickness at the stress point of the blade, this resulted in the maximum deformation being more

than halved.

In terms of load distribution the further improvements were also successful, as seen on the contours the red regions have shrunk, increasing the blade's stress uniformity. This may be the reason for such a large increase in the factor of safety.

The aluminium blade performed exceedingly well in terms of longevity sitting at a minimum life cycle of 19 million which is 19x higher than what was set in the criteria. Unfortunately, the material did not hit the required fundamental frequency, this is likely due to the fact that aluminium is much denser than CFRP and fundamental frequency is inversely proportional to mass.

Both materials had very similar deformation as well but this was expected since their young's modulus was very similar. The deformations recorded both sat in realistic range. For comparison, one paper recorded results of 34 mm and 80 mm [1]

4 Conclusion

In terms of design, one goal in the second iteration was to increase local thickness without increasing the overall blade thickness. For the most part this was done successfully with only an 8% increase in volume. In terms of mass the first design sits at: 0.59541 kg and the second at: 0.64574 kg with a more drastic difference for aluminium at 1.1179 kg. This is important to recognise for material comparison.

Although the longevity of the materials can not be directly compared as a result of differences in calculation, the weight and fundamental frequency can. CFRP allows for lighter weight blades with a greater vibrational resistance making composites a superior material choice for the operating conditions of a prosthetic running blade. In terms of geometry, the second design is clearly superior to the first. The second design has better stress distribution, significantly lower maximum stress and the fundamental frequency is relatively the same.

In terms of choice for a prosthetic leg, the second design made from CFRP is better. Before use Further study would have to be performed on the blades ability to elastically store energy. If it is required for the blade to be made out of aluminium, possibly for environmental reasons, there would have to be further changes in geometry, such as tapering from 9mm rather than 8mm.

A Key limitation with the current analysis of the prosthetic leg revolves around using a quasi static study for a context that involves dynamic impacts, this ignores the effects that inertia may have on the leg which could result in inaccurate or overestimated life cycles and safety factor. Another limitation is the simplification of CFRP to an isotropic material, while comparison to other literature does showcase sane results it may still have an effect.

5 Appendix

5.1 Materials

CFRP data was acquired from a prosthetic running leg research paper [1].

Property	Value	Unit
Density	1.6	gcm^{-3}
Young's Modulus	70	GPa
Poisson's Ratio	0.1	
Shear Modulus	3.1818×10^{10}	Pa
Yield Strength	600	MPa
Ultimate Strength	600	MPa

Table 4: CFRP material data.

5.2 References

References

- [1] Timur Choban Khidir, Mohammed Salam Taha, Hadeel Ali Hasan, and Hakam A Alemarah. Finite element analysis of prosthetic running blades with different designs (i, ii) using epoxy composite material technology. *International Journal of Mechanical and Thermal Engineering*, 6:128–135, 01 2025. pages 3, 10, 11